

# Majorana Modes in the Machine

## Block 2: The Qubit Encoding

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# Outline

- 1 Jordan-Wigner Transform
- 2 Kitaev Hamiltonian in Qubit Language
- 3 Fermion Parity & Topological Degeneracy
- 4 Numerical Verification
- 5 Summary

# Why map fermions to qubits?

**Goal:** simulate the Kitaev chain on a quantum computer.

Quantum computers operate on **qubits** (two-level systems), not directly on fermions.

**Problem:** fermionic operators anticommute, qubit operators on different sites commute.

$$\{c_i, c_j^\dagger\} = \delta_{ij} \quad \text{vs} \quad [\sigma_i^\alpha, \sigma_j^\beta] = 0 \quad (i \neq j)$$

## Jordan-Wigner solution

Attach a **string** of  $Z$  operators to each fermionic operator. The string encodes the fermion occupancy of all preceding sites and restores the correct anticommutation.

# The Jordan-Wigner mapping

$$c_j = \underbrace{\left( \prod_{k=0}^{j-1} Z_k \right)}_{\text{JW string}} \sigma_j^-, \quad c_j^\dagger = \left( \prod_{k=0}^{j-1} Z_k \right) \sigma_j^+ \quad (1)$$

where  $\sigma^\pm = \frac{1}{2}(X \pm iY)$  are the qubit raising/lowering operators,  $X, Y, Z$  are the standard Pauli matrices, and  $Z_k$  acts only on qubit  $k$ .

## Key identities

$$\begin{aligned} c_j^\dagger c_j &= \frac{I + Z_j}{2} \quad (\text{number op.}) \\ c_j^\dagger c_{j+1} + \text{h.c.} &= -\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2} \\ c_j^\dagger c_{j+1}^\dagger + \text{h.c.} &= -\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2} \end{aligned}$$

## Why the string works

$Z_k$  flips sign when site  $k$  is occupied. For  $i < j$ :  $(c_i^\dagger)(c_j)$  picks up  $Z_i \cdots Z_{j-1}$  from the JW string of  $c_j$ , turning commutation into *anticommutation*.

# Substituting JW into $H_K$

Applying the identities to each term of

$$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j (c_j^\dagger c_{j+1} + \text{h.c.}) + \Delta \sum_j (c_j^\dagger c_{j+1}^\dagger + \text{h.c.}):$$

$$H_K = -\frac{\mu}{2} \sum_j Z_j + \frac{t - \Delta}{2} \sum_j X_j X_{j+1} + \frac{t + \Delta}{2} \sum_j Y_j Y_{j+1} \quad (2)$$

## Structure

- $\mu \rightarrow$  **transverse field** ( $Z$  terms)
- Hopping  $t \rightarrow XX + YY$   
(number-conserving)
- Pairing  $\Delta \rightarrow XX - YY$   
(number-breaking)

## Sweet spot $t = \Delta$

The  $XX$  terms cancel.

$$H = -\frac{\mu}{2} \sum_j Z_j + t \sum_j Y_j Y_{j+1}$$

A pure transverse-field +  $YY$  model.

# Fermion parity is conserved

The **fermion parity operator**

$$P = \prod_{j=0}^{L-1} Z_j, \quad P^2 = I, \quad [H_K, P] = 0 \quad (3)$$

splits the  $2^L$ -dimensional Hilbert space into two sectors:

- **Even** parity ( $P = +1$ ): even number of fermions
- **Odd** parity ( $P = -1$ ): odd number of fermions

## Topological degeneracy in the qubit language

In the topological phase the Majorana zero modes hybridize the two parity sectors. The lowest even- and odd-parity states become **degenerate** — the *parity gap*  $|E_0^{(+)} - E_0^{(-)}| \rightarrow 0$ .

This gap is exactly the Majorana hybridization splitting  $E_0(L)$  from Block 1, now visible as a *spectral* signature in qubit space.

# Building the qubit Hamiltonian in code

```
def kitaev_qubit_hamiltonian(L, t=1.0, mu=0.0, delta=1.0):
    dim = 2 ** L
    H = np.zeros((dim, dim), dtype=complex)

    for j in range(L):
        H -= (mu / 2) * pauli_at(Z, j, L)           # transverse field

    for j in range(L - 1):
        H += ((t - delta) / 2) * two_site(X, X, j, L)   # XX coupling
        H += ((t + delta) / 2) * two_site(Y, Y, j, L)   # YY coupling

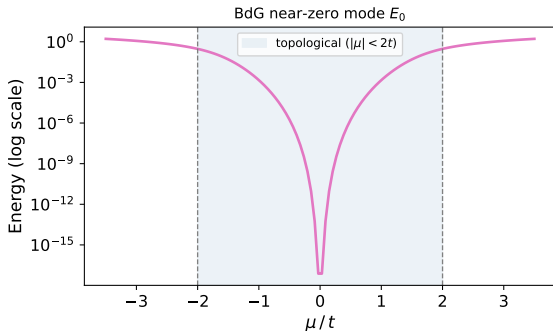
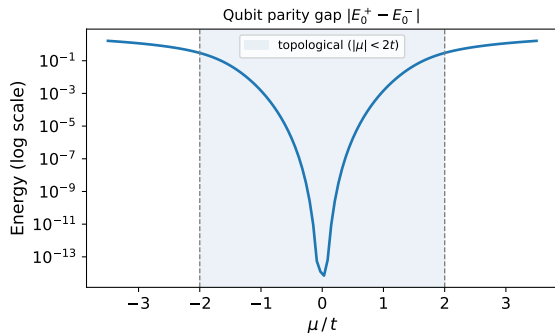
    return H
```

## Tensor-product helpers

`pauli_at(op, j, L)` embeds a single-site Pauli at site  $j$  via  $I \otimes \cdots \otimes \text{op}_j \otimes \cdots \otimes I$ .  
`two_site(A, B, j, L)` places  $A_j \otimes B_{j+1}$ . Both are thin wrappers around `np.kron`.

# Parity gap (qubit) vs BdG splitting — cross-check

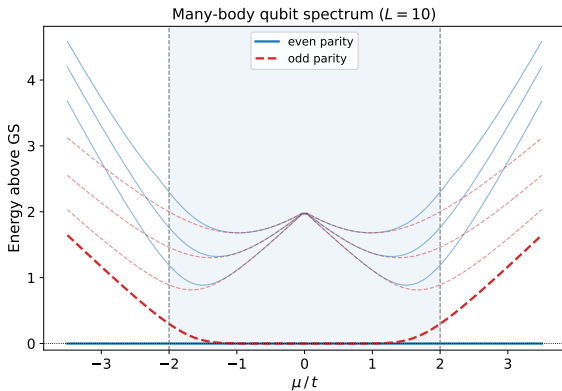
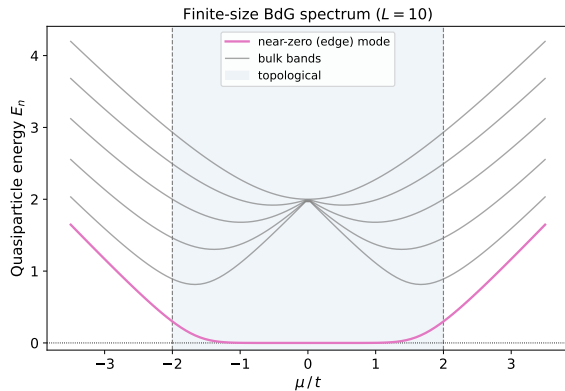
Qubit parity gap vs BdG splitting ( $L = 10$ ,  $t = \Delta = 1$ , OBC)



Both methods give the same exponential decay in the topological phase and the same gap size in the trivial phase — confirming the JW mapping is exact.



# Many-body qubit spectrum vs BdG spectrum



Even (blue) and odd (red) parity sectors are degenerate inside the topological phase and split outside — the bulk-boundary correspondence is manifest directly in the many-body spectrum alongside the BdG modes.

## Block 2 — key takeaways

- 1 The **Jordan-Wigner transform** maps fermions to qubits by attaching a  $Z$ -string that restores anticommutation.
- 2 The Kitaev Hamiltonian becomes an **XY model with transverse field**:  
$$-\frac{\mu}{2} \sum Z_j + \frac{t-\Delta}{2} \sum XX + \frac{t+\Delta}{2} \sum YY.$$
- 3 **Fermion parity**  $P = \prod Z_j$  is conserved — it partitions the spectrum into even/odd sectors.
- 4 The **parity gap**  $|E_0^+ - E_0^-|$  is exponentially small in the topological phase and of order the bulk gap in the trivial phase — a direct qubit-space signature of topology.
- 5 Qubit and BdG spectra **agree exactly** — validating the mapping numerically.

**Next (Block 3):** Measuring topology — circuit-based measurements, edge & string observables.